# Rubin Rao

rubinrrao@tamu.edu | linkedin.com/in/rubinrrao

## Education

**Bachelor of Science in Mechanical Engineering** 3.50 | Aug. 2024 - May 2028
*Texas A&M University* *College Station, TX*

## Objective Statement

Aspiring Mechanical Engineer committed to owning problems end-to-end across the full engineering lifecycle, from conceptualization to analysis and design to manufacturing. Seeking opportunities for the Summer of 2027.

## Experience

**Textron Systems - Uncrewed Land and Air Systems (ULA)** Jun. 2026 – Aug. 2026
*Mechanical Design Engineering | Internship* *Hunt Valley, MD*

- Supported Aerosonde motor characterization test stand development through mechanical design and hardware integration
- Created CAD model and manufacturing drawings for a 15 kW generator test stand using NX
- Developed and released manufacturing drawings for various bracketry and assemblies using GD&T principles

**Bell Flight** Jun. 2025 – Aug. 2025
*Manufacturing/Process Engineering | Internship* *Grand Prairie, TX*

- Designed and iterated a rivet installation tool to resolve recurring quality defects on the 407 Tail Rotor Shaft Assembly
- Reorganized 150+ masking rubber types across two storage units, implementing a labeled catalog to improve traceability
- Collaborated with operators to document production grind setup and procedures across 15+ standard work operations

**Texas A&M Sounding Rocketry Team (SRT)** Aug. 2024 – Aug. 2026
*Structures Engineer | Student Design Team* *College Station, TX*

- Owned the design, analysis, and manufacturing of a modular gantry system capable of withstanding high dynamic loads
- Manufactured structural components using oven-cured preimpregnated composites, layups, and TIG Welding
- Presented and defended structural designs in 3+ formal Design Reviews

## Projects

**Aerosonde Motor Characterization Bracketry** Jun. 2026 – Aug. 2026
*Textron Systems - ULA* *Hunt Valley, MD*

- Designed bracketry to mount sensors and motors to test stand for characterization of thrust, torque, and propeller speeds
- Developed milled aluminum adapter to interface between load cells and 2+ motor configurations
- Architected system to place Hall Effect sensor within 2 mm of encoder face, while withstanding vibrations produced by test stand equipment
- Leveraged Ansys to perform modal & structural analysis on bracketry to characterize & mitigate vibrational & loading effects

**C.R.A.N.E.** Aug. 2025 – Mar. 2026
*Texas A&M Sounding Rocketry Team* *College Station, TX*

- Developed a gantry system and tether system to transport a 400-lb Lander Vehicle, engineered to withstand 2,000 lbs equivalent dynamic loading
- Performed beam bending hand calculations to properly size Gantry System and Tether System hardware
- Conducted Finite Element Analysis (FEA) in Ansys Static Structural to identify potential stress concentrations in the mounting hardware under the 2,000-lb loading condition and drive solutions to redistribute mechanical stresses

**Bell 407 Tail Rotor Blind Rivet Installation Tool** Jun. 2025 – Aug. 2025
*Bell Flight* *Grand Prairie, TX*

- Identified operational root causes of blind-rivet installation defects on the Bell 407 tail rotor shaft
- Designed a tool system to constrain shaft movement and align the rivet gun normal to the rivet head, incorporating technician feedback to improve system ergonomics and accessibility
- Utilized Fusion360 and FDM 3D printing to rapidly iterate through four prototype generations, improving fit, function, and accessibility at each stage

## Technical Skills, Involvement

- **Software:** Ansys (Static Structural, Modal) NX, Solidworks, Fusion 360, Microsoft Office Suite
- **Manufacturing:** Composites (Wet layup, Pre-Impregnated Fiber), TIG Welding, 3D Printing, Laser Cutting (Wood)
- **Involvement:** Texas A&M Sounding Rocketry Team, National Eagle Scout Association